

ELECTRICAL PROPERTIES OF NEURONS

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LEARNING OBJECTIVES

At the end of the lecture, student should be able to

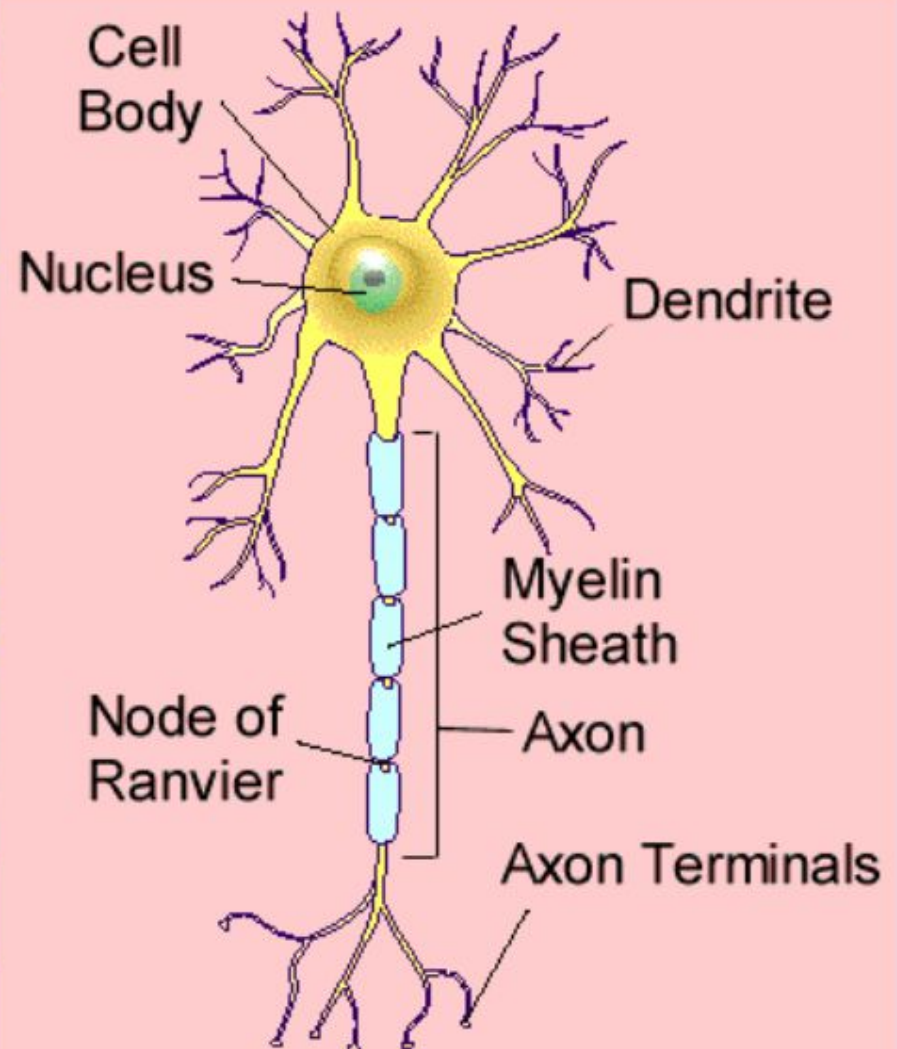
- ✓ understand the morphology of neuron and its basic physical and electrical properties.**
- ✓ Explain Resting Membrane Potential and Action Potential.**
- ✓ Describe Excitation and conduction .**
- ✓ Enlist different Ion Channels.**

NERVOUS TISSUE

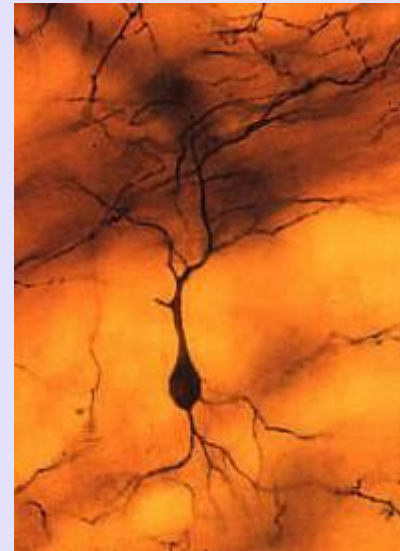
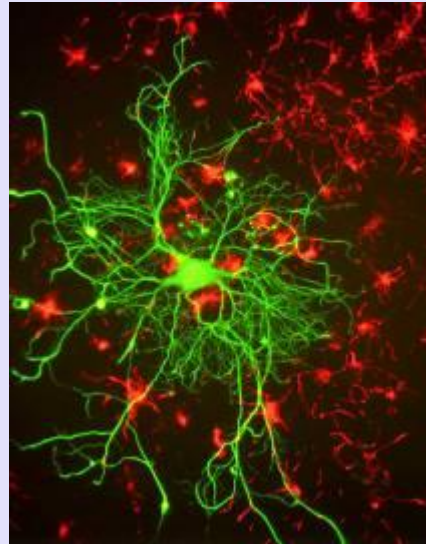
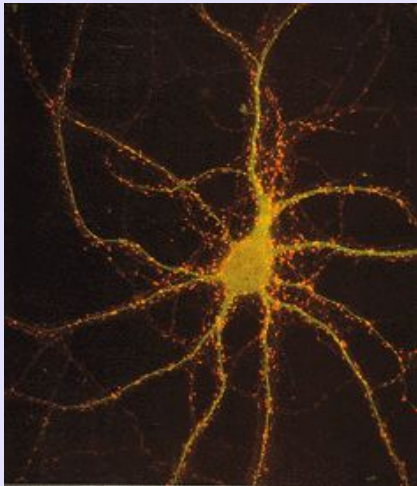
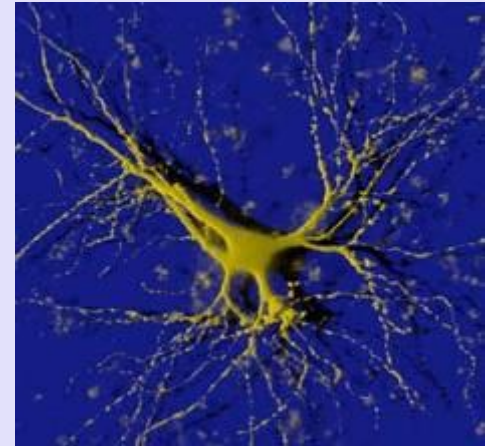
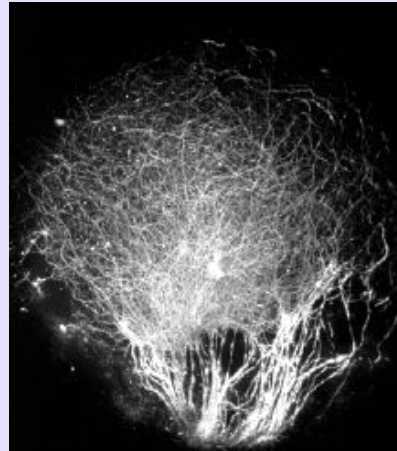
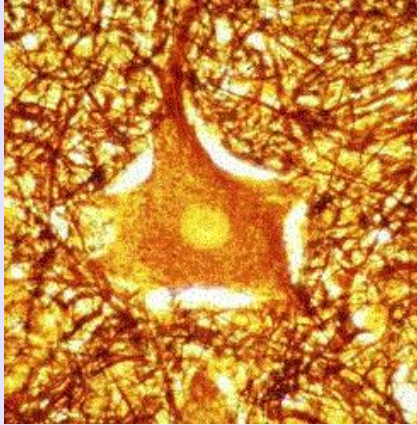
- Nervous Tissue comprises of Neurons, Neuroglia, and blood vessels
- **Neuron** is the basic structural and functional unit of nervous tissue
- **Neuroglia** is the supporting or glial tissue that is 10-times more abundant in mammalian brain than neurons

NEURON

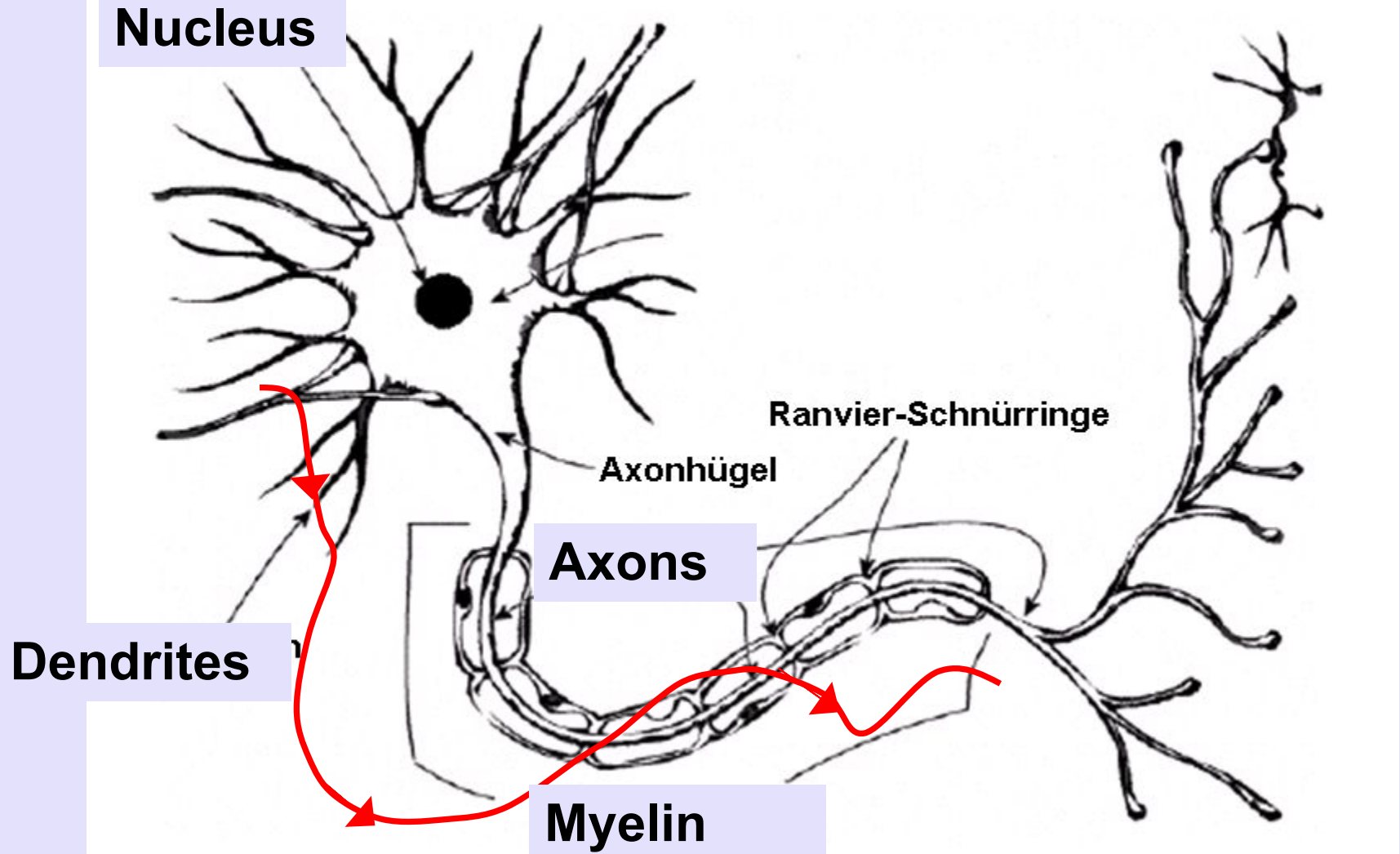
- ❑ Basic structural and functional unit of the nervous system
- ❑ Specialized cells conduct electrical impulses along the plasma membrane
- ❑ Composed of:
 - ❑ **Body:** Soma/Perikaryon
 - Body Contains;*
 - Nucleus:
 - Cytoplasmic organelles
 - Inclusions
 - Cytoskeletal components
 - ❑ **Processes:**
 - Axon and Dendrites



Morphology of Nerve cells (*The Neuron*)

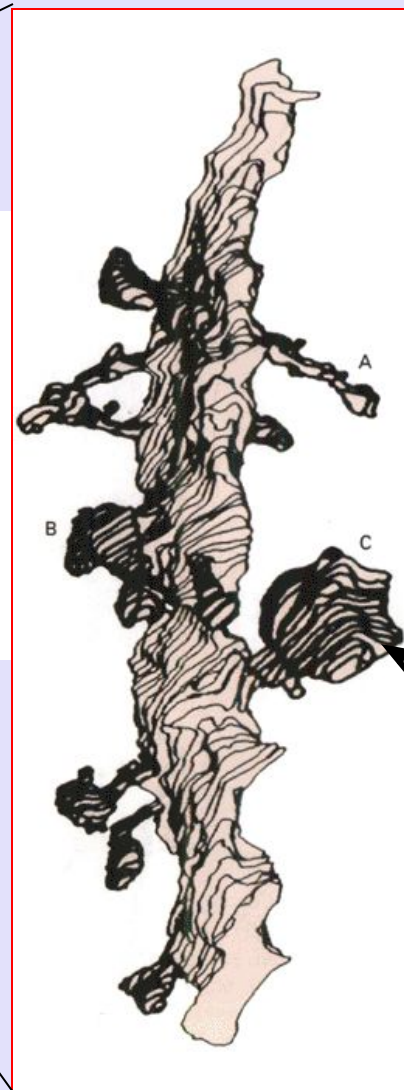
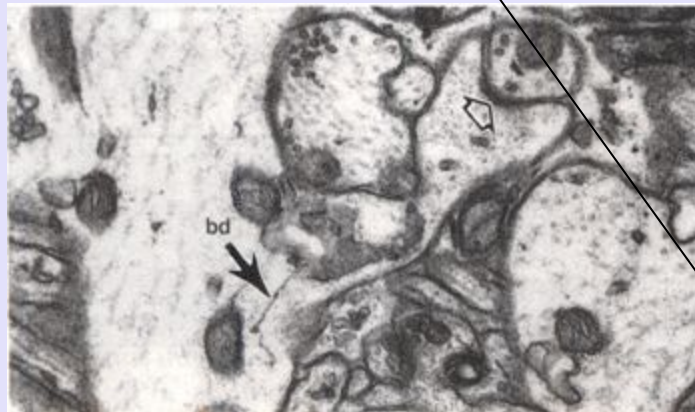
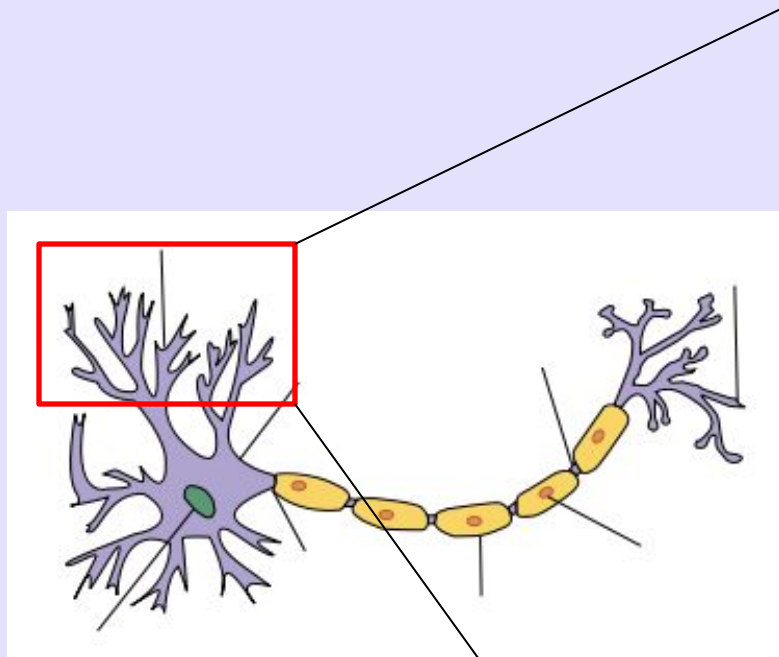


A Neuron



- Dendrites -- Input
- Cell body (soma) -- Integration
- Axon -- Output

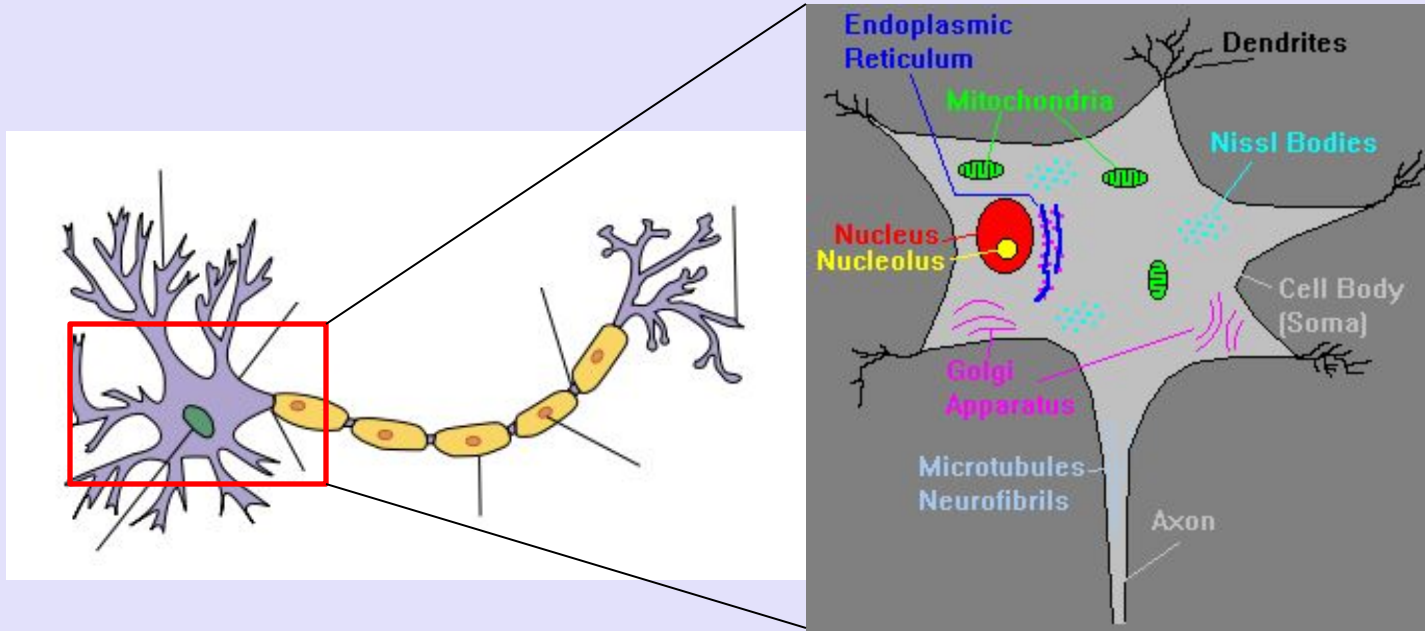
Structure of neurons - Dendrites



At dendrites, the neurons receive input via axons of other neurons at *synapses*

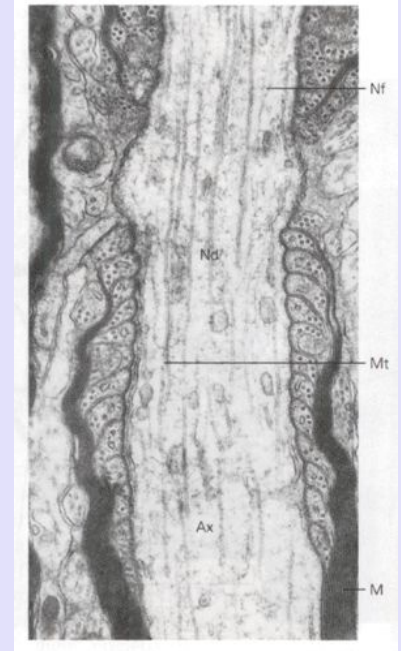
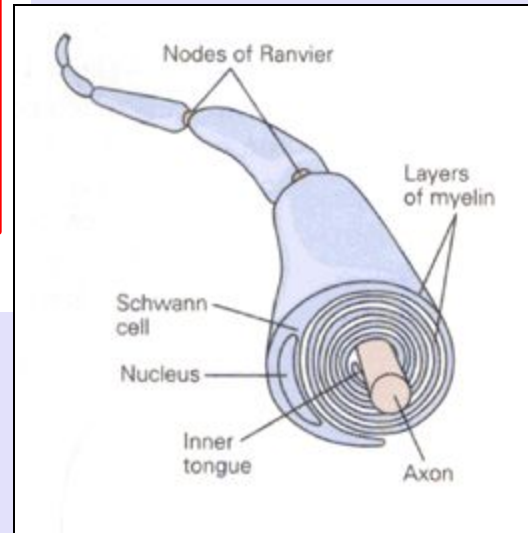
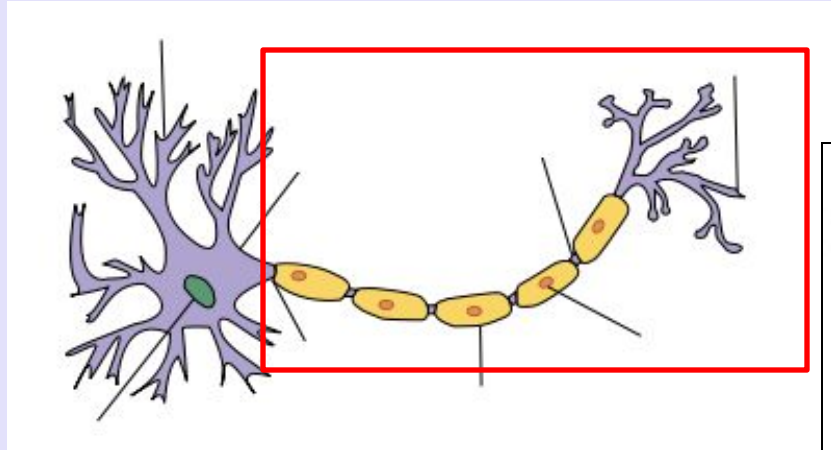
dendritic spine

Structure of neurons - Soma



In the soma of the cells, the *cell nucleus* is located (containing the DNA, i.e. genetic code); the *synthesis of the proteins* (within ribosomes and endoplasmatic reticulum) as well as *energy production* (mitochondria) are performed.

Structure of neurons - Axon



**The axon transmits the information electrically from the soma to the synapses –
it is surrounded by myelin that insulate the axon,
provided by oligodendrocytes (glial cells)**

NEURON: MORPHOLOGICAL CLASSIFICATION

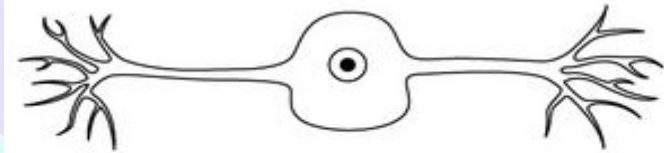


- Unipolar
- Bipolar
- Multipolar

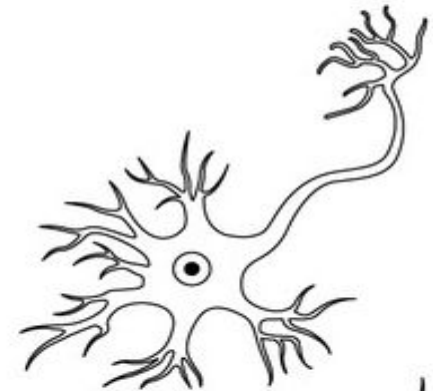
- Single axon & multiple dendrites
- Most common type in men, e.g
- Motor cortex, interneurons, ...

- Pseudounipolar

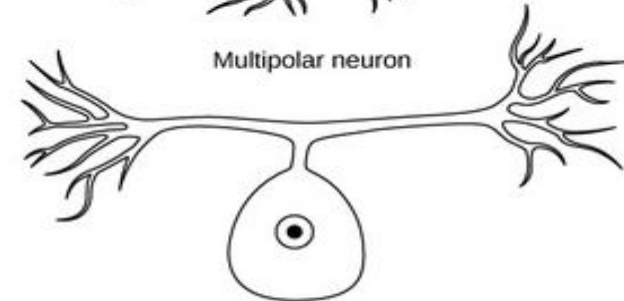
- Single process arises from body
- Branches into an axon and dendrite, e.g
- Present in spinal and cranial ganglia



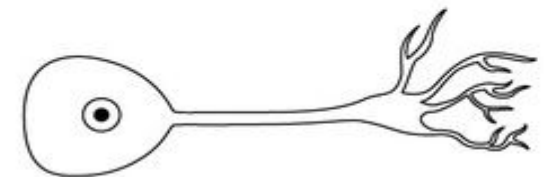
Bipolar neuron



Multipolar neuron

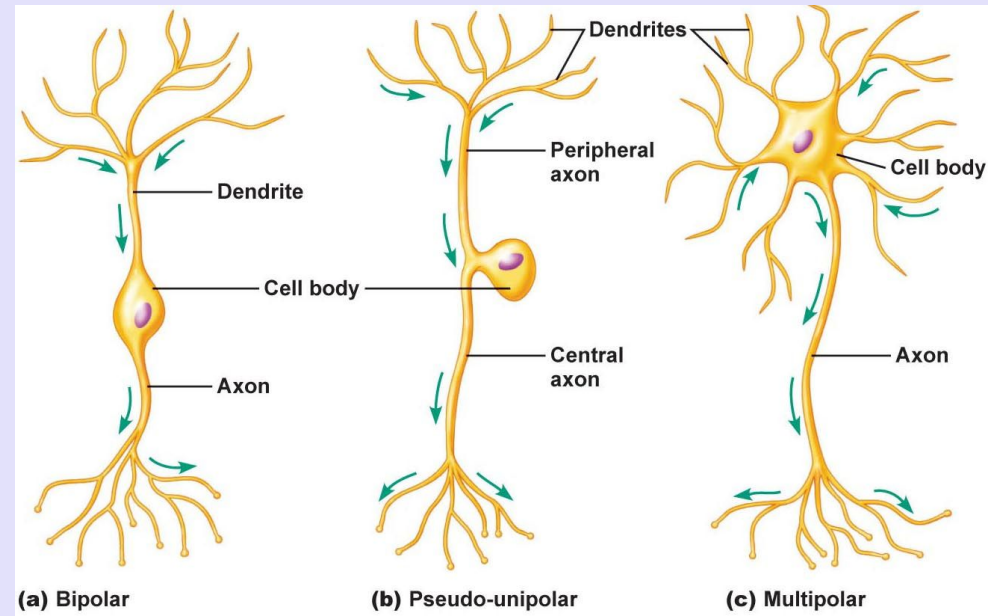
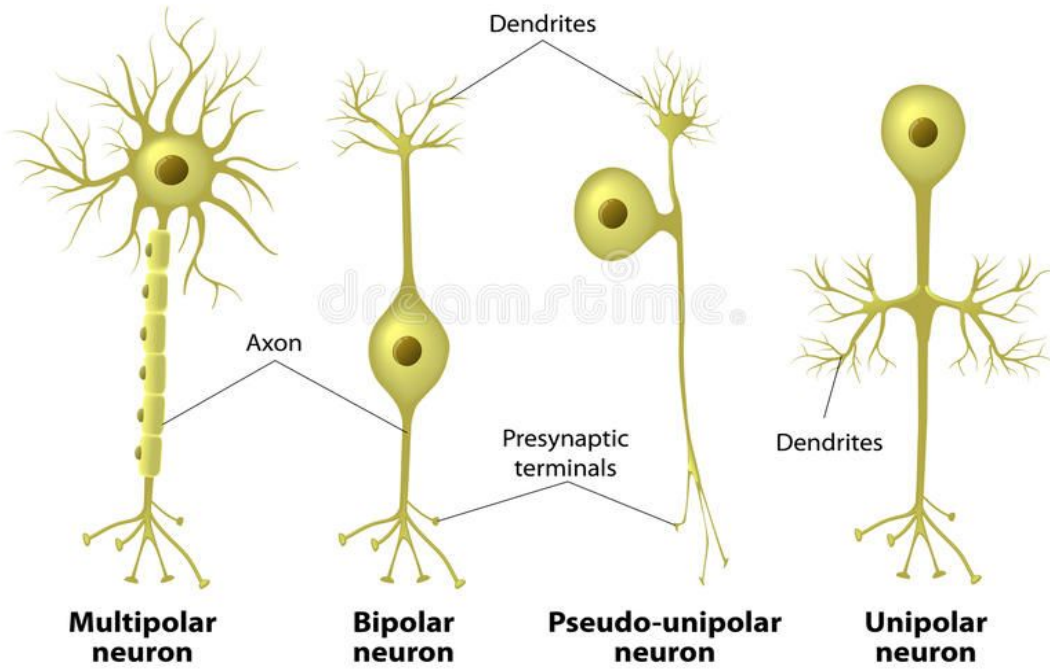


Pseudounipolar neuron

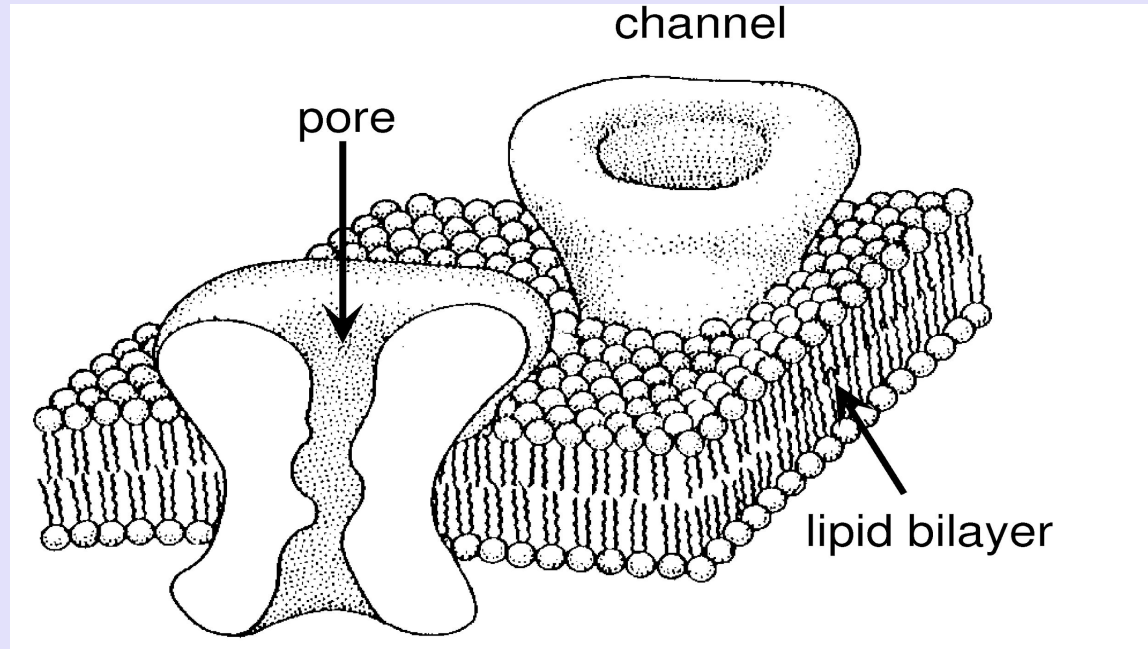


Unipolar neuron

Basic Neuron Types



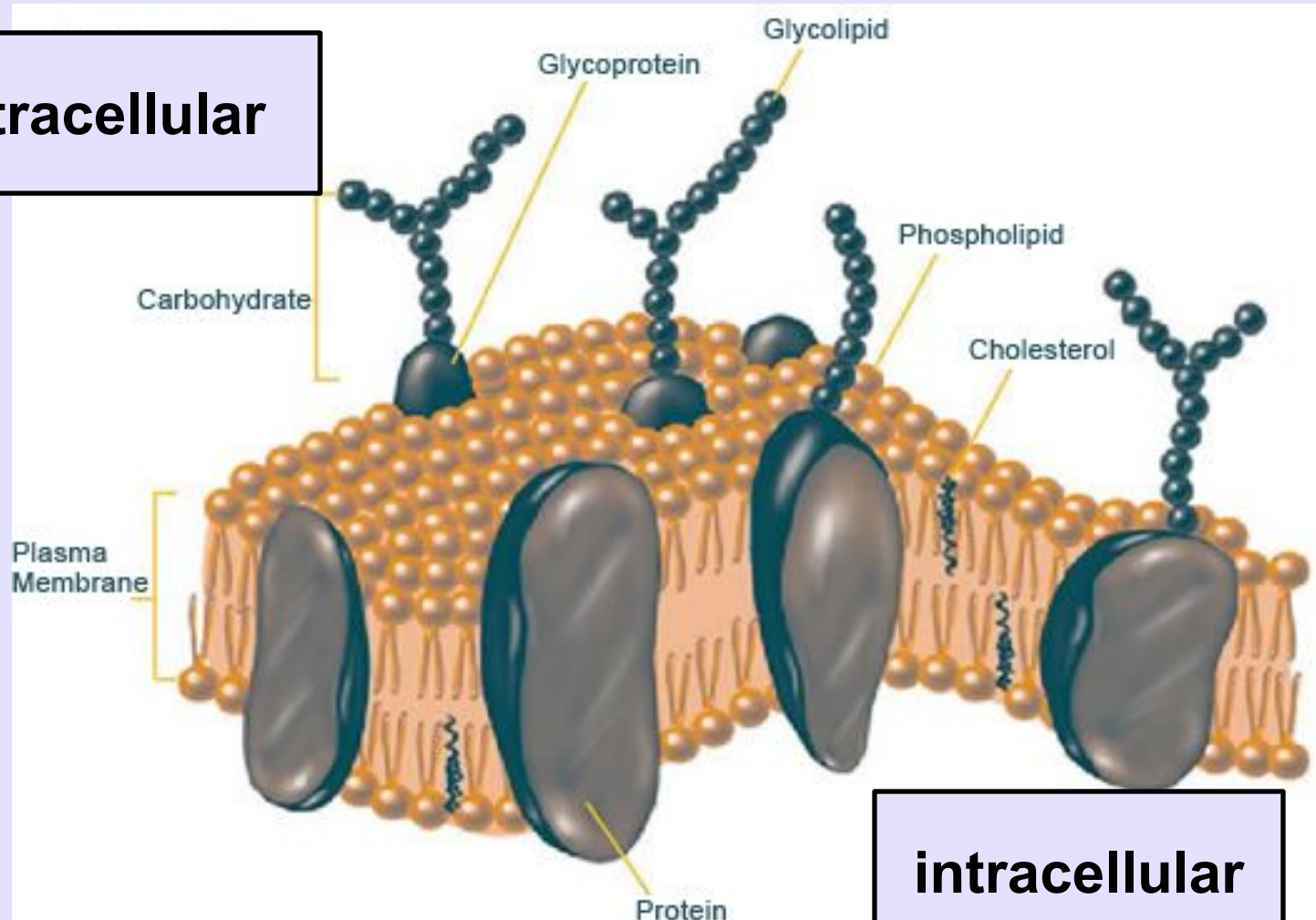
Electrical Properties of Neuron



- Neurons are enclosed by a membrane separating interior from extra cellular space
- The concentration of ions inside is different (more –ve) to that in the surrounding liquid

Electrical properties of neurons

extracellular

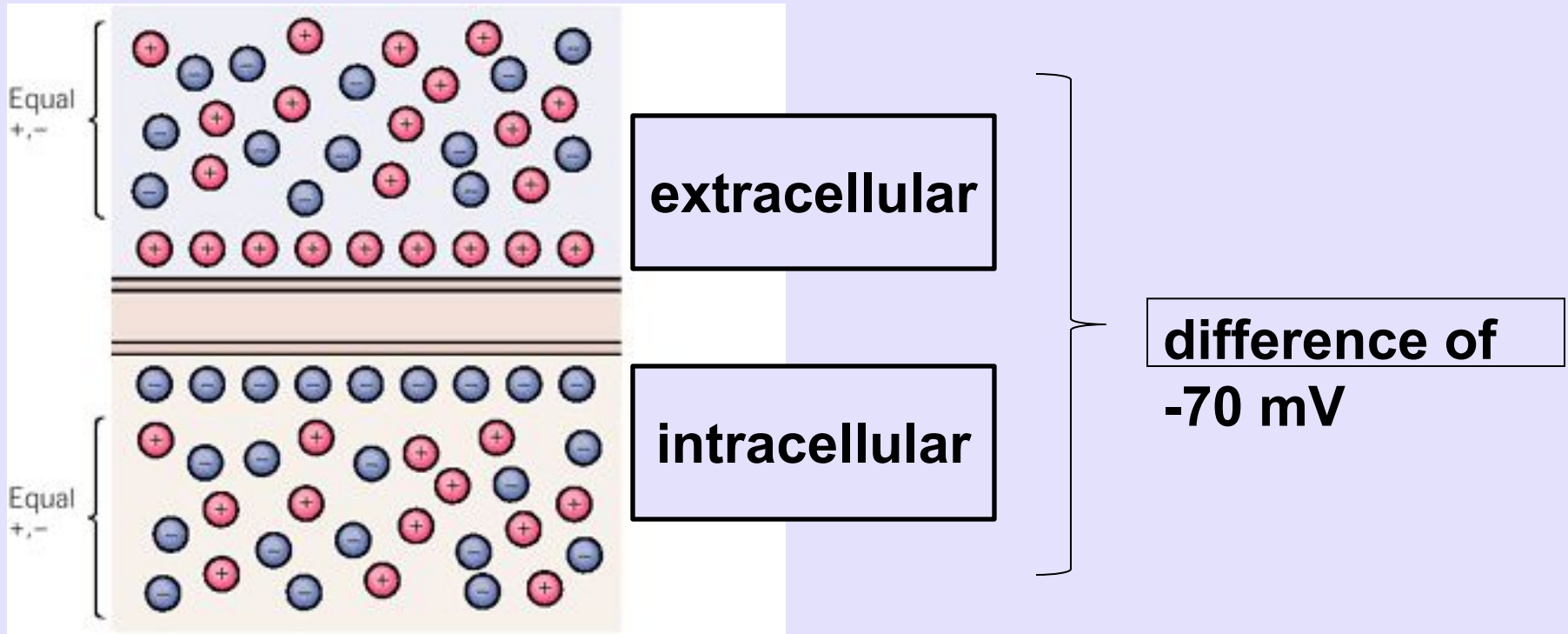


intracellular

The cell membrane isolates the intracellular from extracellular space

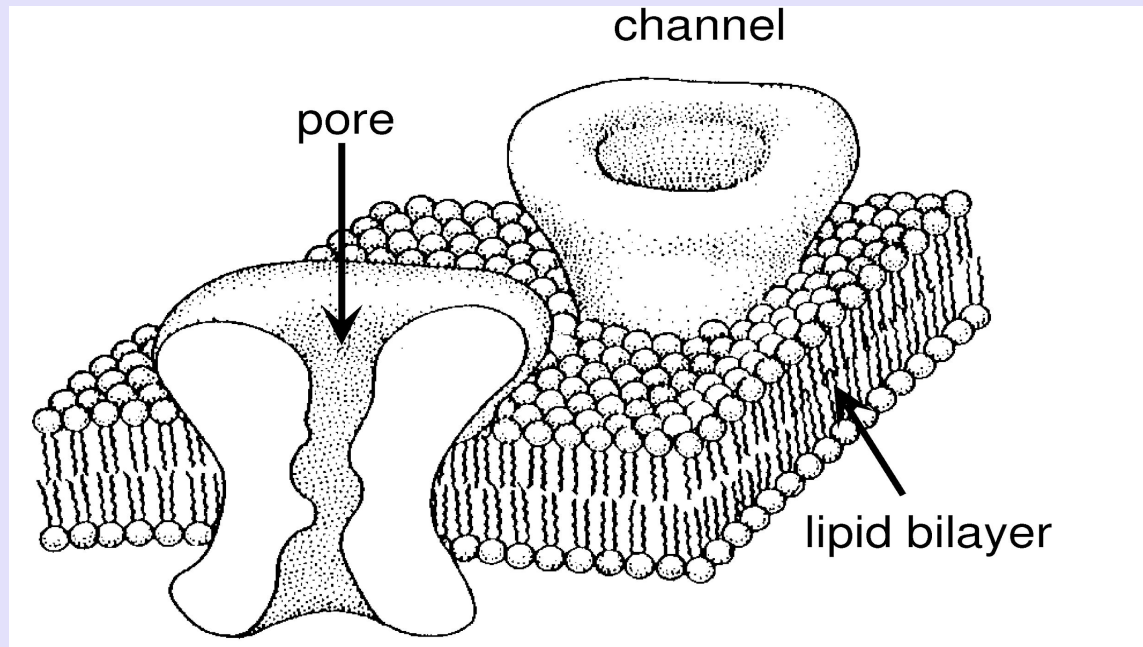
Electrical properties of neurons

The membrane potential



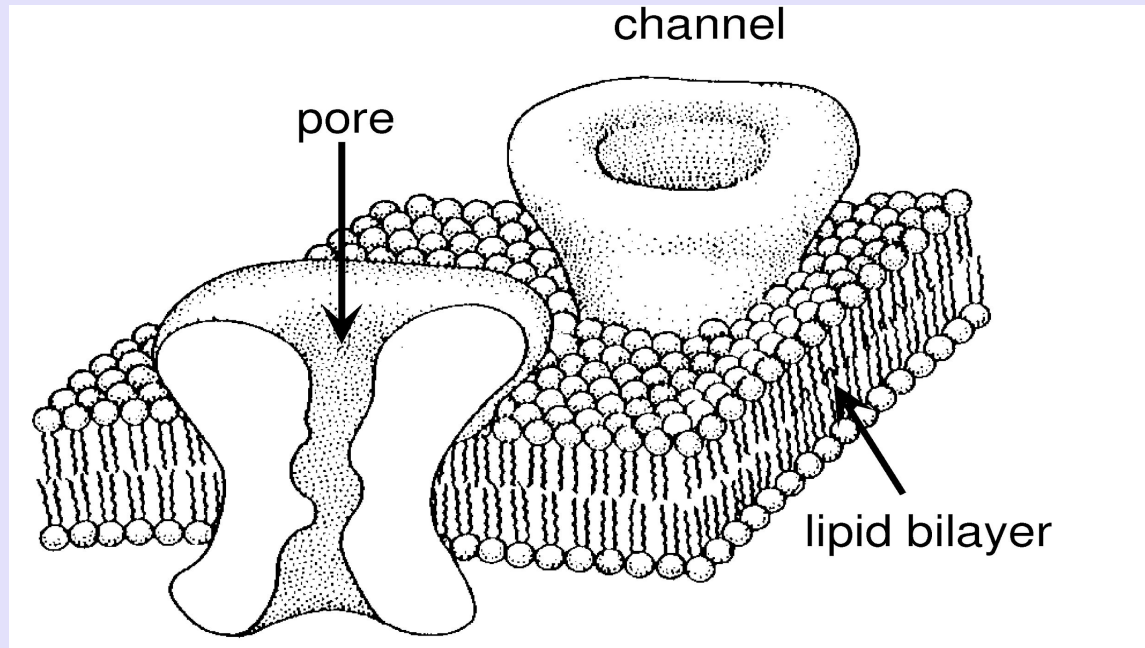
In the resting state, the intracellular space contains more negative ions than the extracellular space

Electrical Properties of Neuron



- Cell membrane: 2-3 nm thick and is impermeable to most charged molecules and so acts as a capacitor by separating the charges lying on either side of the membrane.
- NB Capacitors, store charge across an insulating medium. Don't allow current to flow across, but charge can be redistributed on each side leading to current flow.

Electrical Properties of Neuron



- The ion channels are proteins in the membrane, which lower the effective membrane resistance.

Resting membrane potential

- All cells (not just excitable cells) have a **resting potential (membrane potential)** : an electrical charge across the plasma membrane
“ A potential difference across the cell membrane at the resting stage or when the cell is not stimulated ”
- When a neuron is at rest, the inside of the neuron is negative relative to the outside.
- It is constant or stable
- It is different in different cells

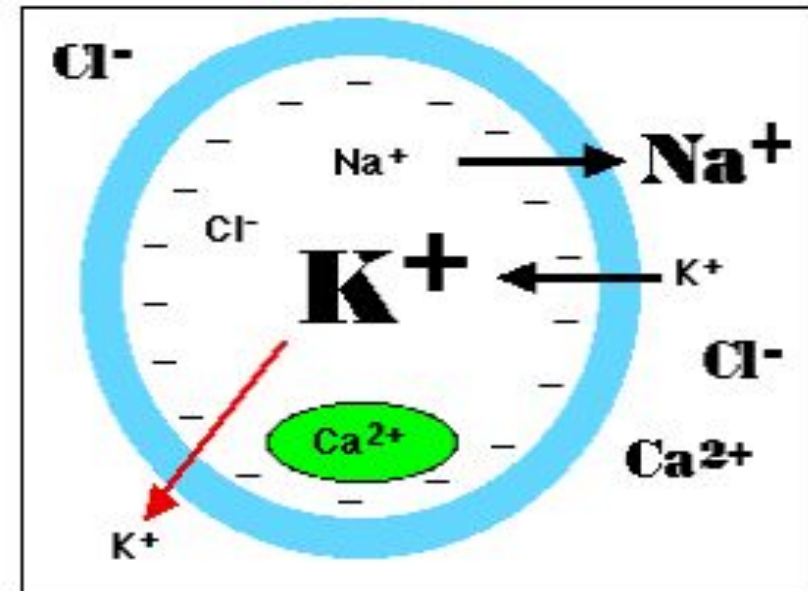
RESTING MEMBRANE POTENTIAL

- The resting membrane potential is the potential difference that exists across the membrane of excitable cells such as nerve and muscle in the period between action potentials (i.e., at rest).
- The resting membrane potential is established by diffusion potentials, which result from the concentration differences for various ions across the cell membrane. Each permeant ion attempts to drive the membrane potential toward its own equilibrium potential. Ions with the highest permeabilities or conductances at rest will make the greatest contributions to the resting membrane potential, and those with the lowest permeabilities will make little or no contribution.

Resting membrane potential

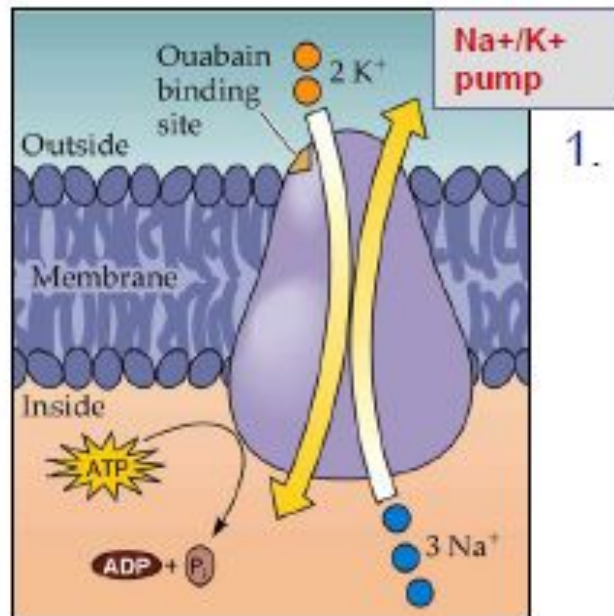
Unequal distribution of ions across the plasma membrane

- $[Na^+]$ outside $10\times >$ inside
- $[K^+]$ inside $20\times >$ outside
- $[Cl^-]$ and $[Ca^{2+}]$ outside $>$ inside
- $[Ca^{2+}]$ high in some intracellular compartments (green oval)



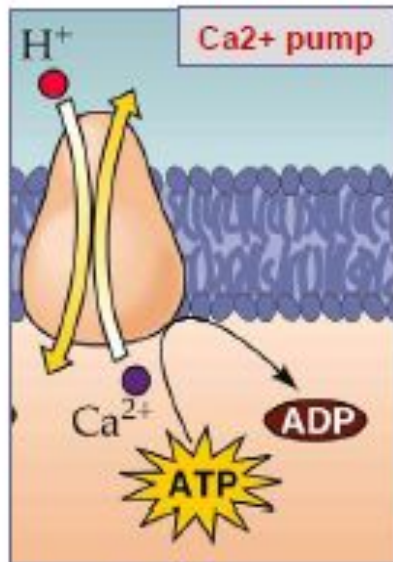
Extracellular and Intracellular Ion Concentrations		
Ion	Concentration (mM)	
	Intracellular	Extracellular
Mammalian neuron		
Potassium (K^+)	140	5
Sodium (Na^+)	5–15	145
Chloride (Cl^-)	4–30	110
Calcium (Ca^{2+})	0.0001	1–2

Resting membrane potential

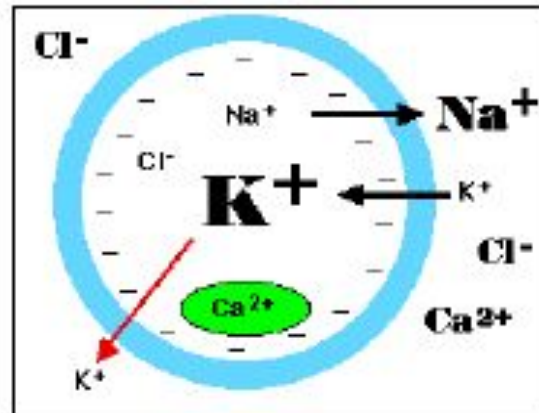


1. The Na⁺/K⁺ transporter (ATPase/pump) - active transport

- Two K⁺ into the cell
- Three Na⁺ out of the cell
- Result: a net loss of positive charges within the cell



2. Some **potassium channels** (non-gated) in the plasma membrane are "leaky" allowing a slow **facilitated diffusion** of K⁺ out of the cell (red arrow).



3. Ca²⁺ is removed via Ca²⁺ pump

What sets the membrane potential ?

- **Each ion gradient** contributes to the total membrane potential.

- ❑ **Na⁺ contributes +58mV**
- ❑ **K⁺ contributes – 75mV**
- ❑ **Cl⁻ contributes -65mV**
- ❑ **Ca⁺² contributes +155mV, etc**

As a result:

- ❑ **The interior of the cell negative**
- ❑ **Typically -70mV** (varies -40 to -90 mV depends upon the type of neuron)

- ✓ Difference in ion concentration between compartments gives rise to the resting membrane potential (RMP).
- ✓ Membrane permeability to these ions influences the RMP.

Excitation and Conduction

- Influx and out flux of ions through the ion channels embedded in membrane, causes generation of action potentials.
- These action potentials or spikes or nerve impulses are the source of communication in most biological neurons.

Action Potential:

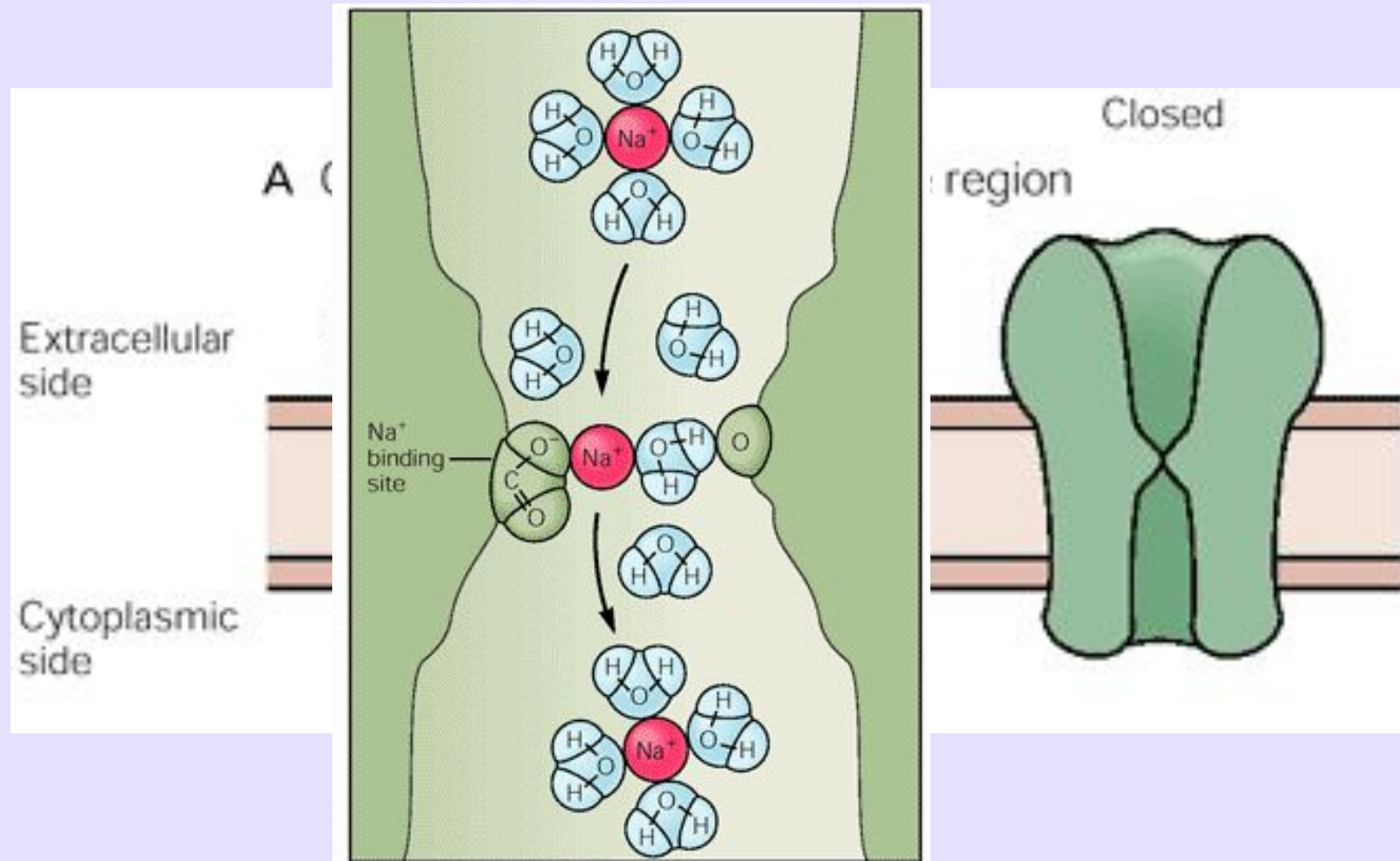
The electrical activity developed in an excitable cell when stimulated

Excitation and Conduction

- An action potential happens when some event happens that stimulates the neuron.
- Nerve cells have a low threshold for excitation – may easily be excited by electrical, chemical or mechanical stimuli
- Two types of physiochemical disturbances are produced as a result
- **Local or Non-Propagated Potentials**
such as Synaptic, Generator or Electrotonic Potentials
- **Propagated Disturbances**
such as Action Potentials or Nerve Impulses

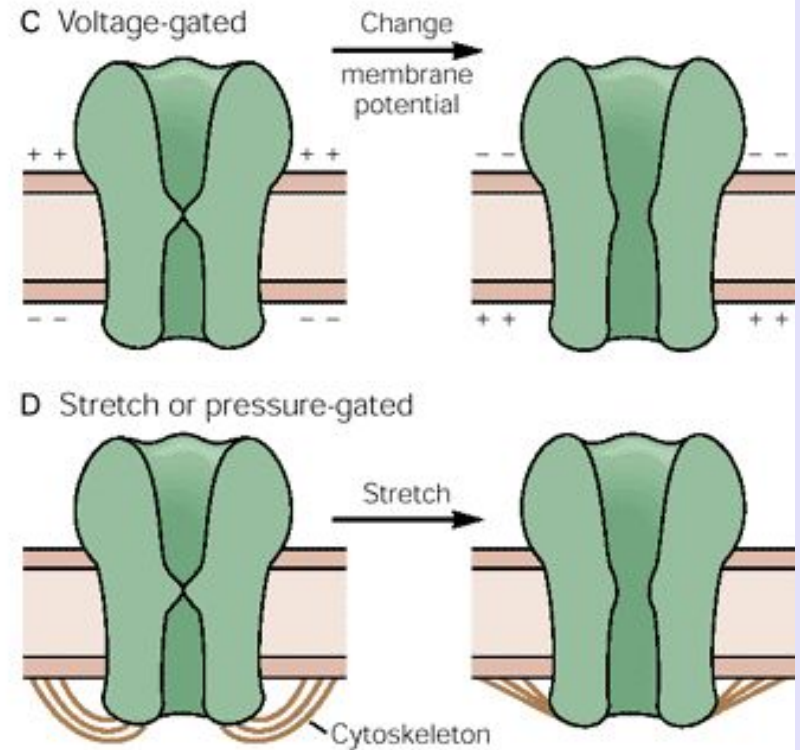
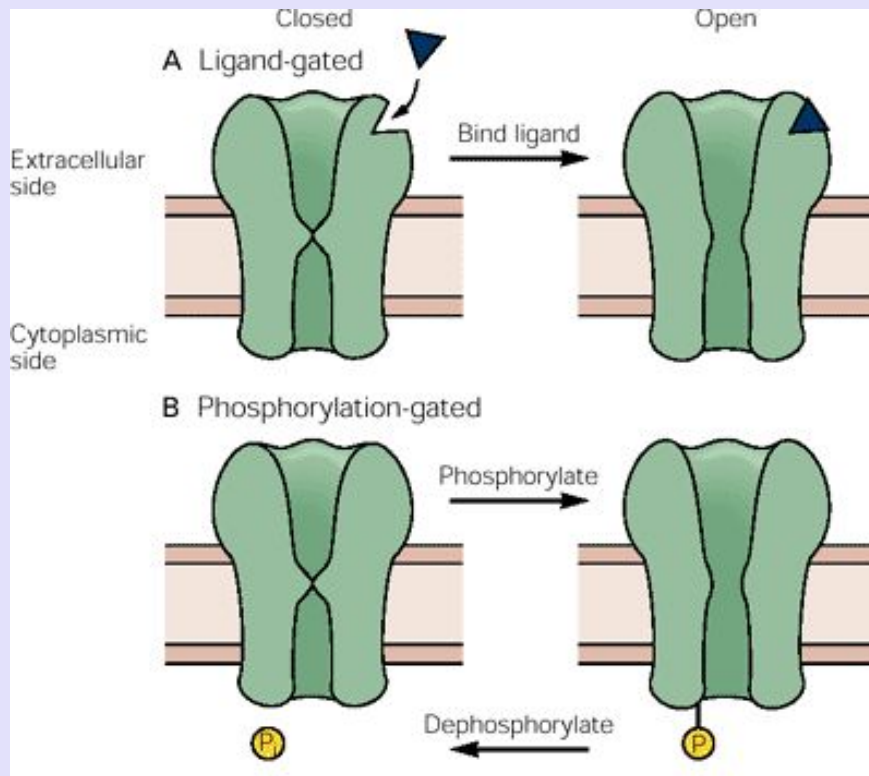
Suitable electrical probe (electrode) and measurement instrumentation (amplifier and read-out) can measure these tiny potentials on the order of few milli-volts

Ion channels connect the intra- and extracellular space



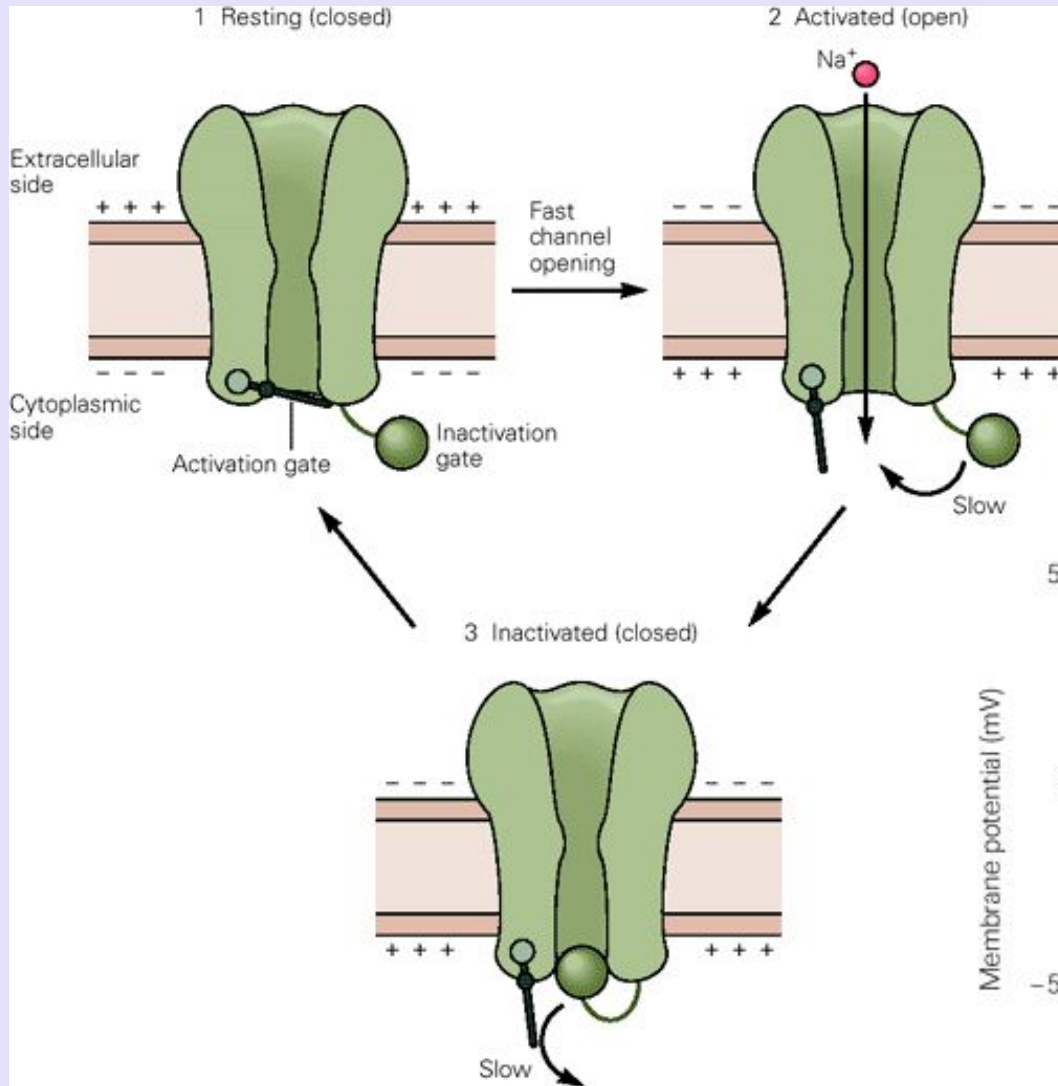
Opening of ion channels lead to a flux of ions through the membrane and to a change of the membrane potential

Different types of ion channels

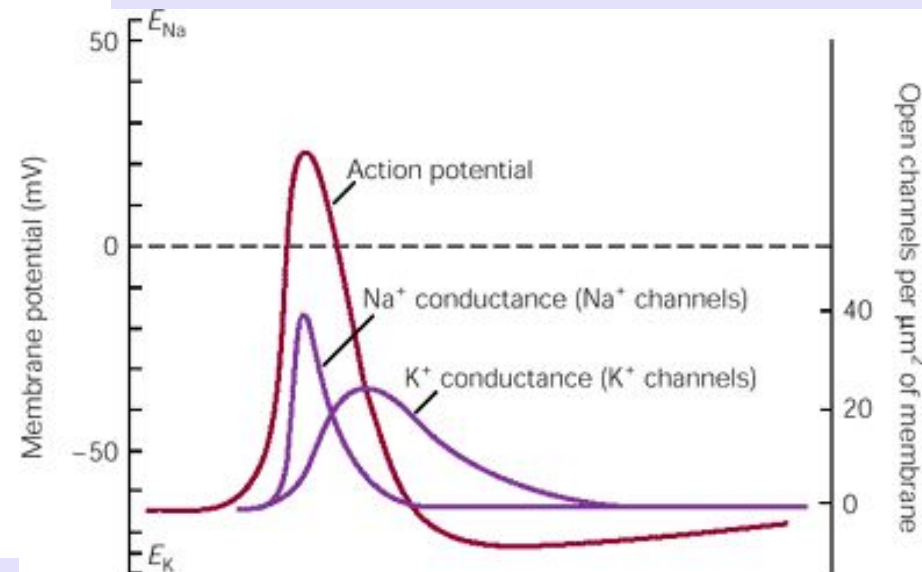


The action potential

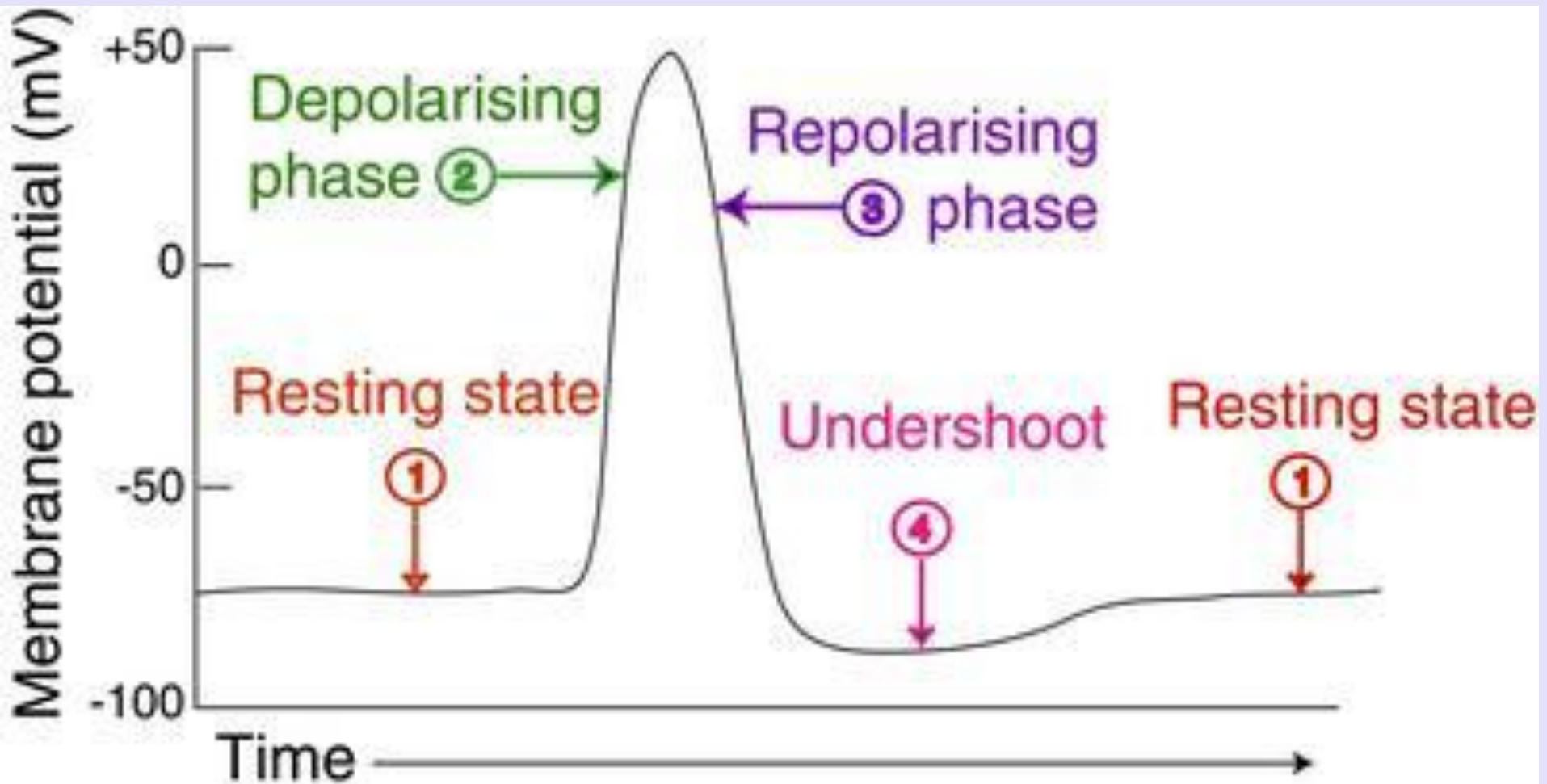
The action potential is generated by ion flux through voltage gated channels



All or nothing principle!!



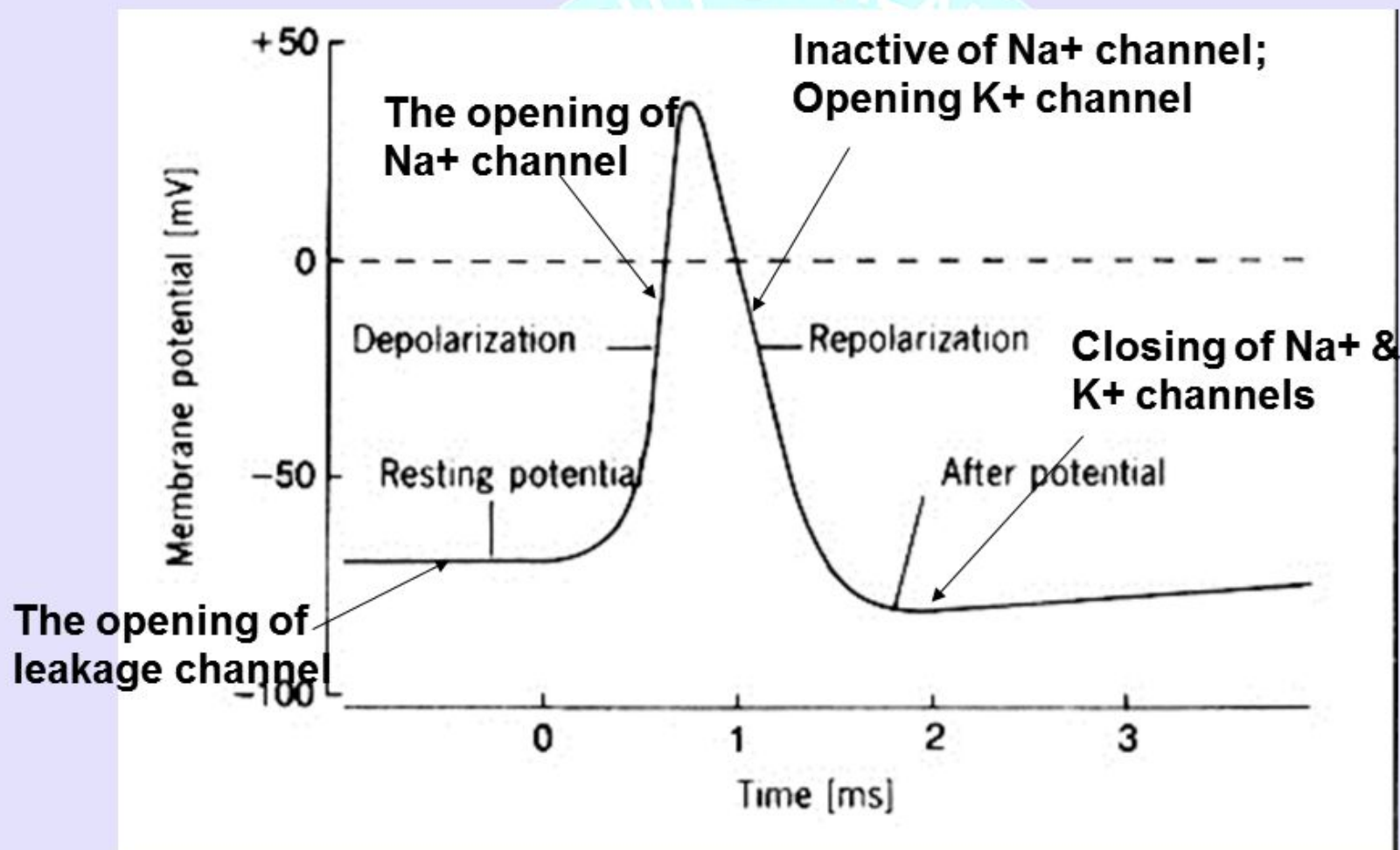
Phases of Action potential



Mechanism of Action Potential

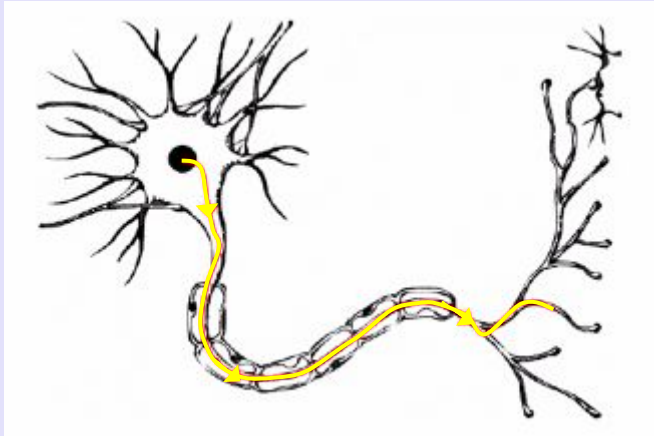
- Action potentials are caused by an exchange of ions across the neuron membrane.
- A stimulus first causes sodium channels to open.
- Because there are many more sodium ions on the outside, and the inside of the neuron is negative relative to the outside, sodium ions rush into the neuron.
- Remember, sodium has a positive charge, so the neuron becomes more positive and becomes depolarized.

CHANNELS DURING ACTION POTENTIAL

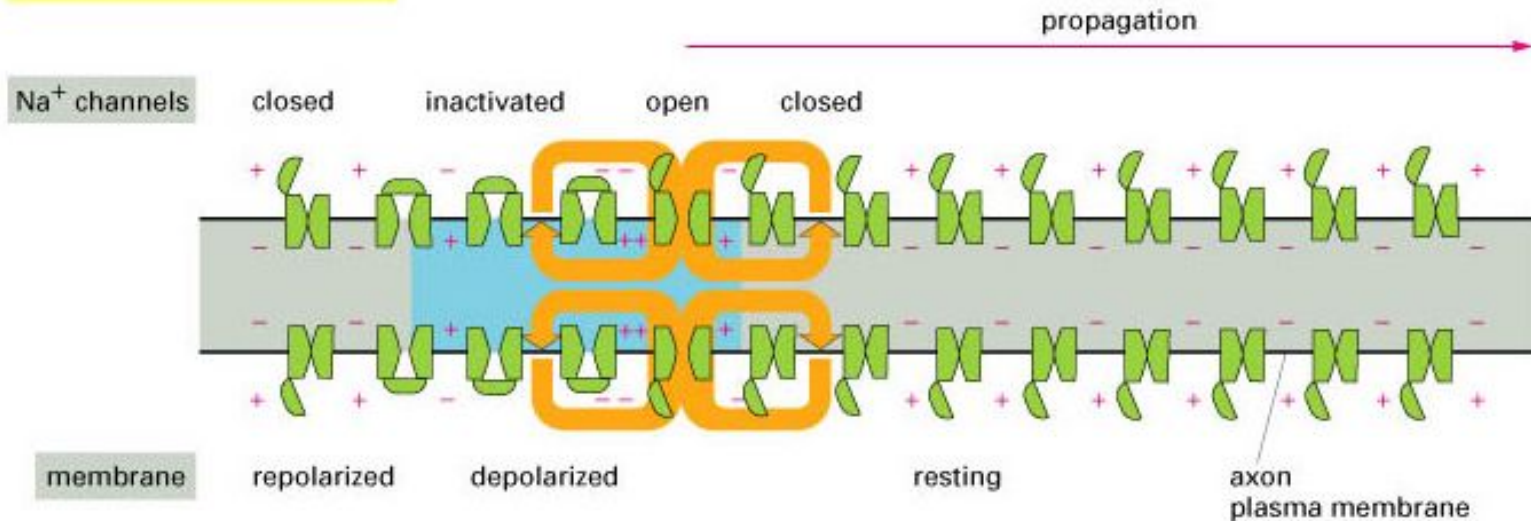


- It takes longer for potassium channels to open.
- When they do open, potassium rushes out of the cell, reversing the depolarization.
- Also at about this time, sodium channels start to close. This causes the action potential to go back toward -70 mV (a repolarization).
- The action potential actually goes beyond -70 mV (a hyperpolarization) because the potassium channels stay open a bit too long.
- Gradually, the ion concentrations go back to resting levels and the cell returns to -70 mV

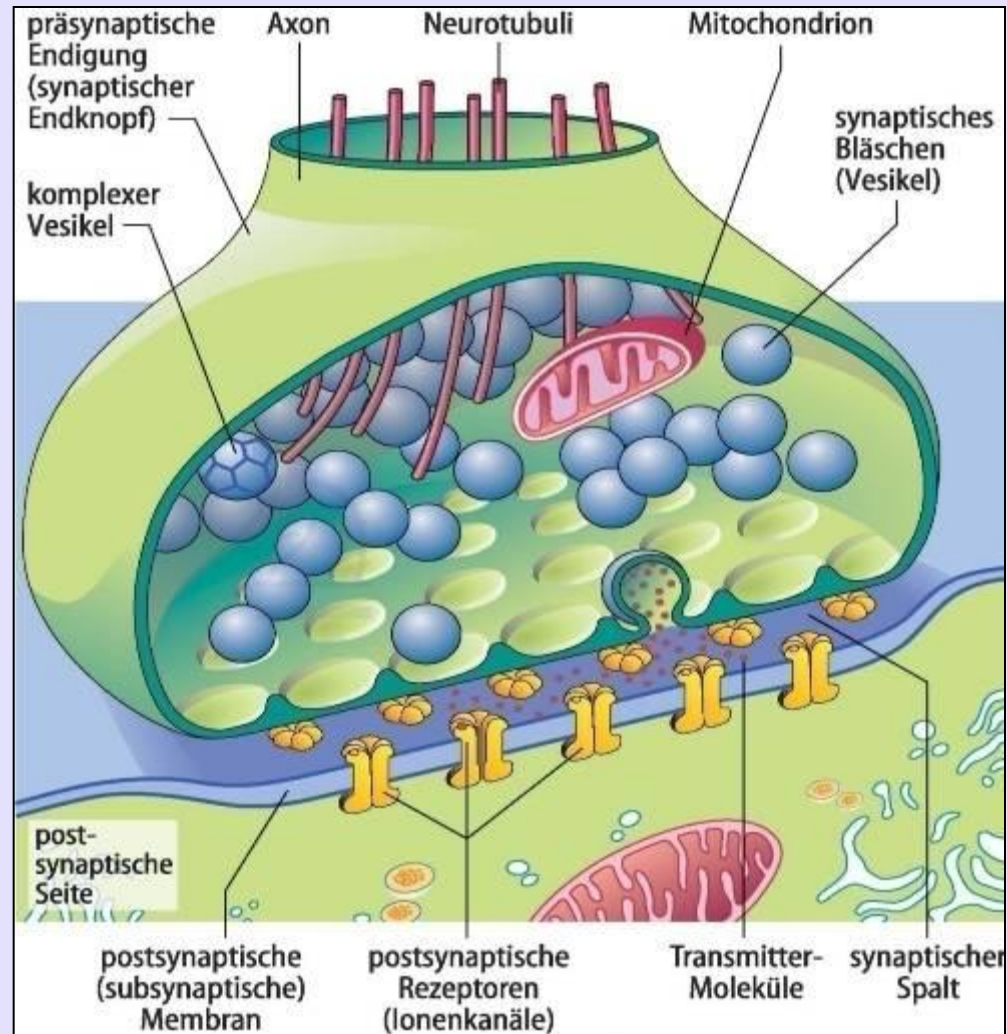
Propagation of the action potential



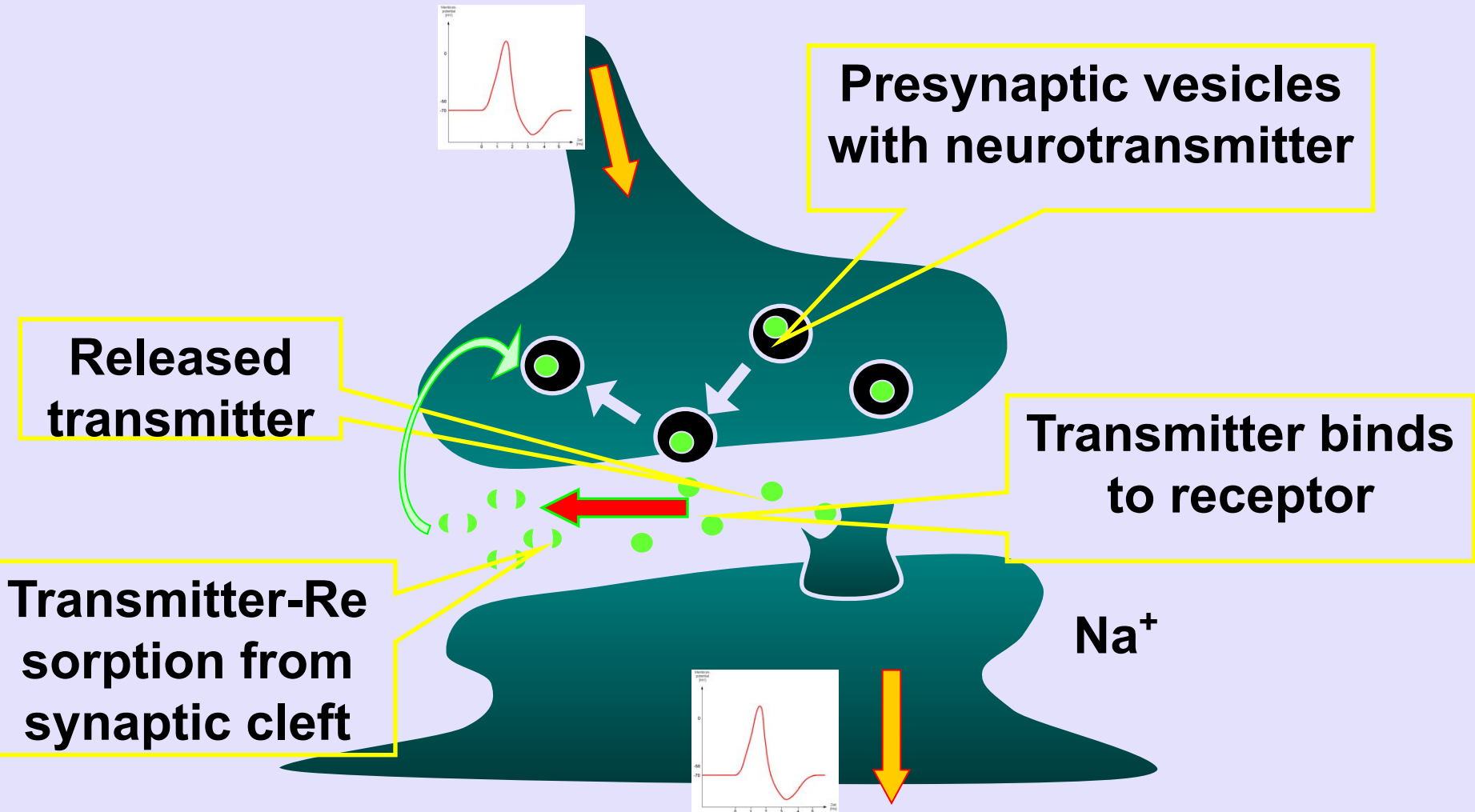
(B) instantaneous view at $t = 0$



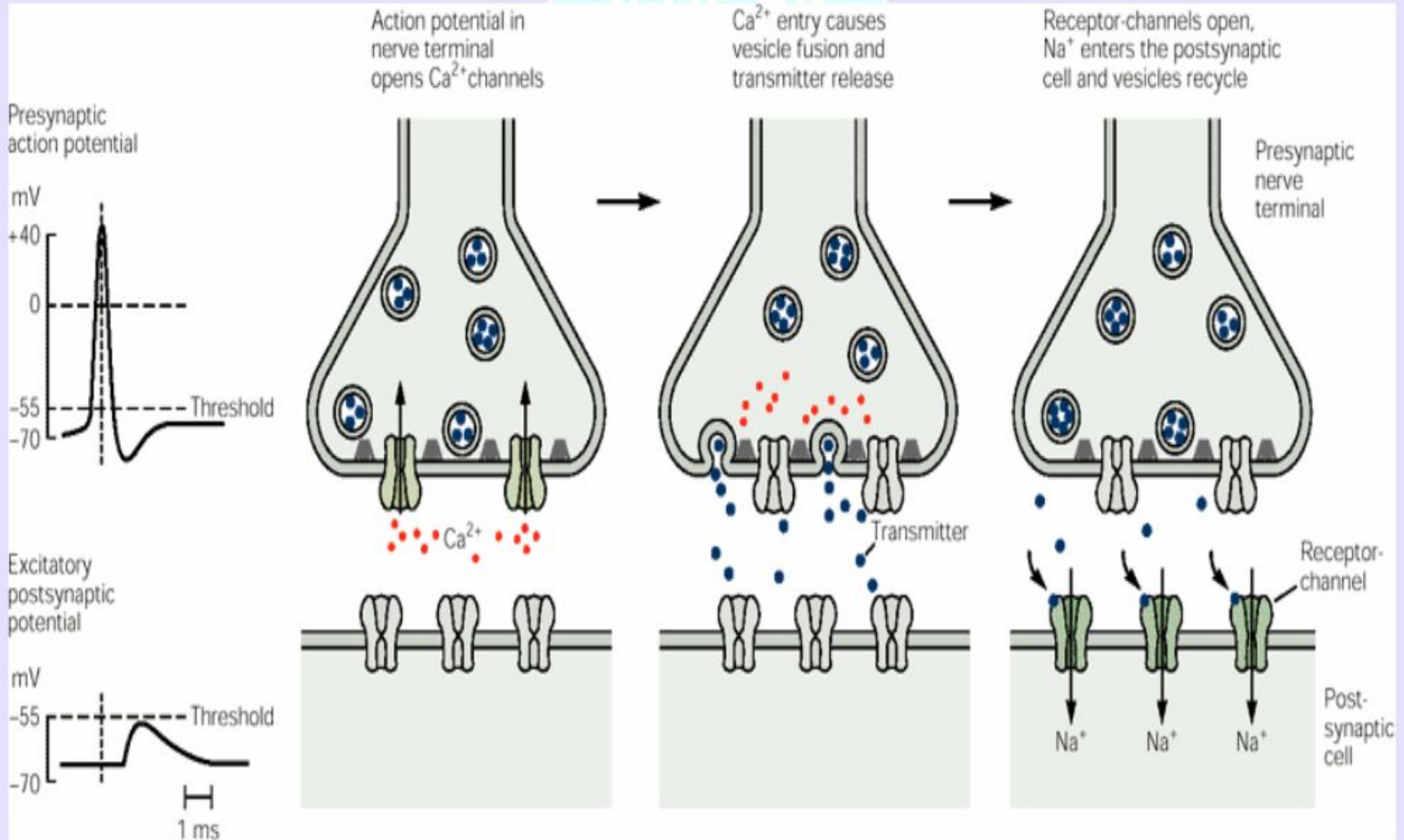
Synapse – Communication between neurons



Synapse – Communication between neurons



SYNAPSE – COMMUNICATION BETWEEN NEURONS



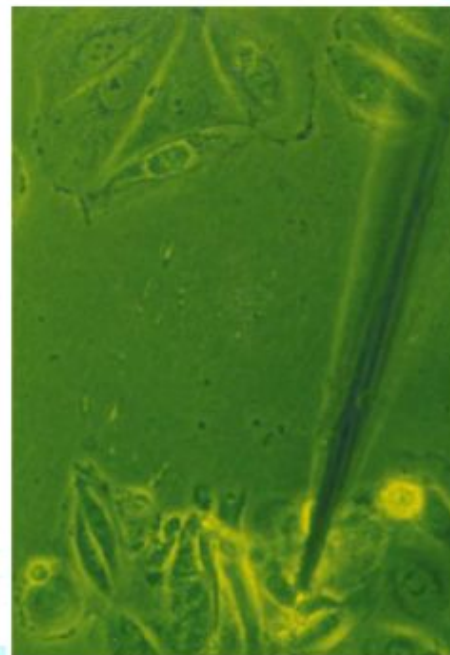
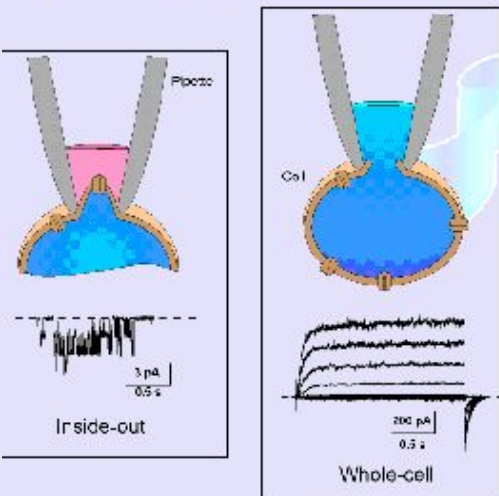
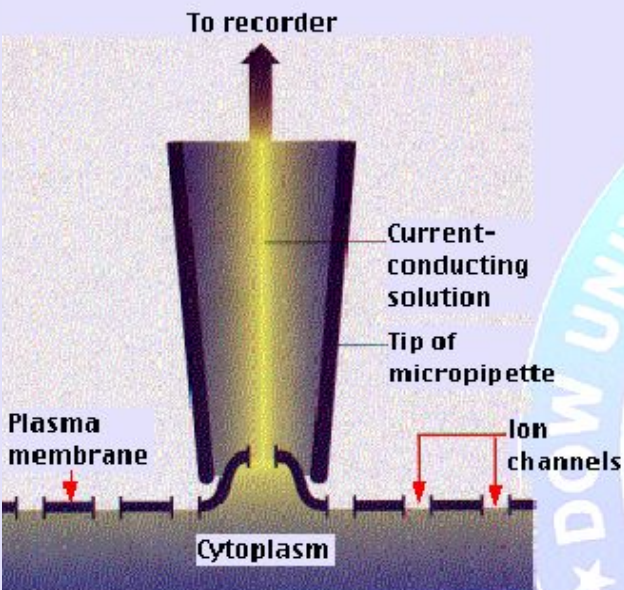
PROPERTIES OF THE ACTION POTENTIAL

- “All or none” phenomenon
A threshold or suprathreshold stimulus applied to a single nerve fiber always initiate the same action potential with constant **amplitude, time course and propagation velocity.**
- Propagation
- Transmitted in both direction in a nerve fiber

The Patch-Clamp Technique

- This is a novel technique (developed by Neher and Sakmann et al. for which they were awarded with a Nobel prize) in which physiological currents flowing through the cells can be detected without disrupting the cell or its contents
- A micropipette (diameter in microns) filled with a buffer solution and carrying a metal electrode is gently touched to the cell membrane to record change in voltage, with the help of Ohm's law.
- In this case the current passing across the membrane is controlled by the experimenter and the resulting changes in voltage are recorded, generally in the form of action potentials.

THE PATCH CLAMP TECHNIQUE



Good point: an optical nanofiber can probe a cell's molecular contents without killing it.

